

FIN — Release 10.22

Nadsoliton Puzzle Atlas

Z12 Simulator Audit, Controlled Cross-Disciplinary Intuition,
and a New Memory Object after Reduction

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Central result. Exact elimination of hidden strict Z_{12} nodes generates a frequency-dependent self-energy and a process with memory. The static Schur complement is only a resolvent limit, not the exact generator of the complete reduced dynamics.

Boundary. The result does not generate a selector, physical units, a legacy-strict bridge, or experimental validation.

Repository. <https://github.com/hyconiek/Fractal-Nadsoliton-Theory>

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Confidence convention

- **[Proven]** — an algebraic or numerical result with an explicit test and tolerance.
- **[Strong evidence]** — stable in the declared test class, but not a general theorem.
- **[Moderate evidence]** — a meaningful mechanism with unresolved identifiability.
- **[Speculative]** — a research hypothesis, not a claim about nature.
- **[Refuted]** — a claim falsified in the declared model.

Chapter 1

Executive verdict

The most productive "assembly of the puzzles" does not yield a new scalar parameter or a hidden selector. It yields the operator-valued object

$$\Sigma_E(z) = A_{EH}(zI + A_{HH})^{-1}A_{HE},$$

the frequency-dependent **self-energy of the reduced process**, obtained after partitioning the nodes into observed E and hidden H sectors. **[Proven]**

This object unifies five previously separate pictures:

1. Green function/resolvent, through the exact block-resolvent formula;
2. Schur compression, as the static limit $\Sigma_E(0)$;
3. wave and diffusion dynamics, through the same analytic resolvent family;
4. memory and environment, through an exact time-domain memory kernel;
5. the observer, as an explicitly supplied accessible/hidden partition rather than a metaphysical component of the equation.

It does not solve the three principal omissions of FIN: it does not select an orientation, generate a physical unit, or provide an experimental preparation and apparatus. **[Proven]**

The deepest intuition supplied by the atlas is therefore:

Fractal reduction should not be modeled solely as a static transformation of one kernel into another. Exact elimination of an information level generates a process with memory. The static Schur complement is only its zero-frequency shadow.

This is a known reduction mechanism in mathematical physics, but its concrete form for the strict FIN operator and its relation to wave–diffusion duality provide a new testable direction inside the project. It is not evidence of new physics.

Chapter 2

Method: intuition with safeguards

"Intuitive" comparison can easily turn graphical resemblance into false equivalence. The atlas therefore uses four layers:

1. **Shape layer:** identify similar plots, symmetries, flows, and diagrams.
2. **Mechanism layer:** determine whether both structures use the same algebraic operation.
3. **Prediction layer:** ask whether the analogy produces a new computable and falsifiable quantity.
4. **Assumption-debt layer:** list everything that must be added before the analogy becomes a physical model.

Each match receives one of five classifications:

- **exact equivalence** — an explicit operation-preserving map exists;
- **common mechanism** — the operation is shared but the objects or interpretations differ;
- **structural analogy** — similar geometry or spectrum without an equivalence theorem;
- **metaphor** — visually useful, with no evidential force;
- **anti-analogy** — the resemblance exposes the precise point at which FIN differs from the comparison theory.

The script `fin_nadsoliton_puzzle_atlas.py` scanned:

- 12 individual characteristics against 15 fields: **180 comparisons**;
- all unordered pairs of characteristics: $\binom{12}{2} = 66$;
- every pair against 15 fields: **990 comparisons**.

This yields **1170 explicit similarity indices**. Tag-based scoring is only a candidate-search device. It is not evidence and never raises the confidence status of a hypothesis.

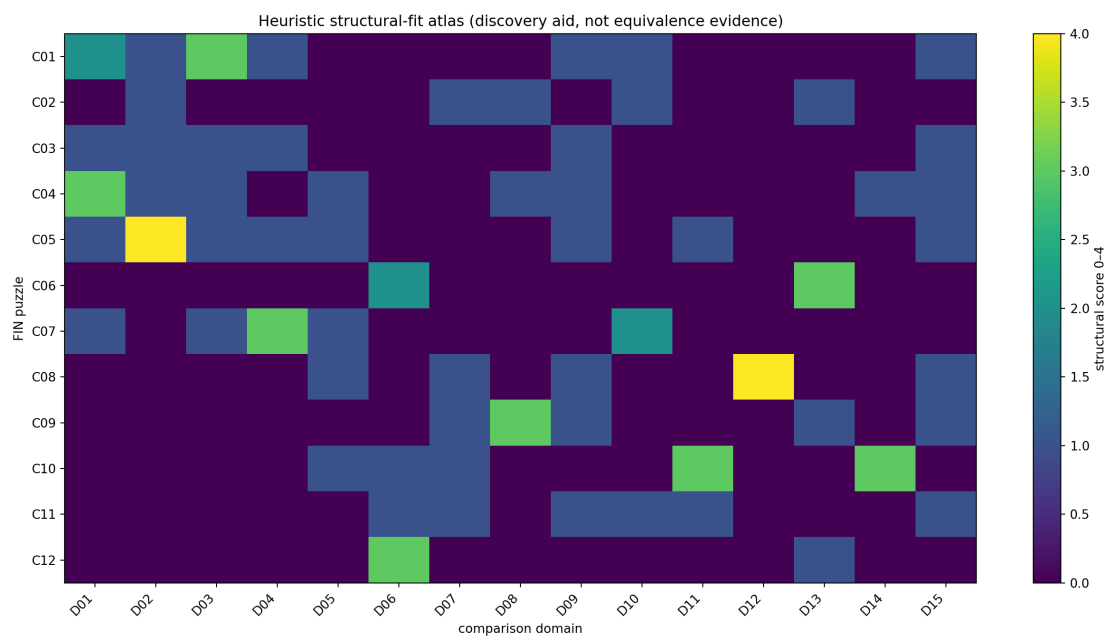


Figure 2.1: Heuristic matching matrix for 12 puzzles and 15 fields. Scores are used only to search for analogies.

Chapter 3

What Z12 sim.html actually does

The code implements a finite thought experiment on Z_{12} :

$$W_{xy} = K(d(x, y)), \quad A = sI - W, \quad U_t = e^{-itA}.$$

The initial state is a partially coherent preparation on two nodes:

$$\rho_0(\phi, \eta) = \frac{1}{2} \left(|a\rangle\langle a| + |b\rangle\langle b| + \eta e^{-i\phi} |a\rangle\langle b| + \eta e^{i\phi} |b\rangle\langle a| \right).$$

After the evolution $\rho_t = U_t \rho_0 U_t^\dagger$, the code evaluates

$$p_x(t) = (\rho_t)_{xx}, \quad J(x \rightarrow y) = 2W_{xy} \operatorname{Im}(\rho_{yx}),$$

and two different aggregate current observables:

$$C_1 = \sum_x J(x \rightarrow x+1),$$
$$C_\chi = \sum_x \sum_{d=1}^5 d J(x \rightarrow x+d).$$

C_1 sees only the first hop harmonic. C_χ is an oriented group current of the full kernel, excluding the antipodal edge $d = 6$, because a sign cannot be assigned to the shortest displacement there without an additional convention.

The simulator's intended distinction is sound: a **sign receiver** is not a **sign source**. A radial operator can transport prepared chirality into a measurable current, but it does not choose the sign of the phase.

Chapter 4

Methodological errors and corrections

4.1 Inconsistent current sign

Two diagonal contributions used the opposite sign of the imaginary part from the interference contributions, violating the single definition of $J(x \rightarrow y)$.

Correction: every contribution is now computed as

$$2W_{xy} \operatorname{Im}(\rho_{yx}).$$

[Proven]

4.2 False claim for $\eta = 0$

Removing coherence eliminates cross terms but need not eliminate the local currents of the two independently propagated packets. The audit gives

$$\max_{x,y} |J_{xy}|_{\eta=0} = 0.0995050916, \quad C_\chi|_{\eta=0} \approx 0.$$

Correction: the interface now states that interference circulation vanishes, not that all local currents vanish. [Proven]

4.3 Confusing one harmonic with the full current

For the preparation $a = 10, b = 2$,

$$C_1(\phi) \approx 0$$

for every tested phase, even though the full operator contains long-range hops. At $\phi = 0.7$,

$$C_\chi = +0.1211880893, \quad C_\chi(-0.7) = -0.1211880893,$$

while

$$C_1(0.7) = -2.36 \times 10^{-16}.$$

Correction: the simulator displays C_χ as the principal counter and C_1 as a blind-harmonic control. [\[Proven\]](#)

4.4 Noncanonical "circulation over all edges"

The original sum weighted distant edges by an implicitly chosen cyclic displacement without declaring the branch choice or antipodal convention.

Correction: the observable is now explicitly defined as C_χ , with a statement that it depends on a chosen orientation and lift of distance. It is a chirality detector, not its source.

4.5 False "asymmetry" from changing separation

The pair $a = 10, b = 3$ still has a reflection interchanging the two equal sources. The test at $\phi = 0$ gives $C_\chi \approx 0$.

Correction: the switch is labeled "different separation," not symmetry breaking. [\[Proven\]](#)

4.6 False monotonicity in $|\phi|$

C_χ is odd and periodic:

$$C_\chi(-\phi) = -C_\chi(\phi), \quad C_\chi(0) = C_\chi(\pi) = 0.$$

It may grow linearly near zero, but not globally with $|\phi|$.

Correction: the narration and plot now describe periodic behavior. [\[Proven\]](#)

4.7 Spectrum–kernel mismatch

The spectrum of A was pasted as a fixed array, so a change of K could leave stale eigenvalues.

Correction: the spectrum is now computed from the kernel by Fourier transform, and the page verifies

$$s = 1.660307278766099, \quad \lambda_{\min}(A) \approx 0, \quad A \succeq 0.$$

4.8 Missing continuity-equation control

Correction: the page checks normalization and

$$\dot{p}_x + \sum_y J(x \rightarrow y) = 0.$$

The independent NumPy audit gives a maximum residual of 5.55×10^{-17} .

4.9 Misleading visual elements

The central arrow rotated even at zero current, while the gauge scale depended on interaction history.

Correction: zero current is now static, and the scale is determined reproducibly from the current $C_\chi(\phi)$ curve.

4.10 Audit verdict

audit_z12_sim.py passes **15/15 tests**. **[Proven]**

This does not show that FIN is physically realized. It establishes only the internal consistency of the declared finite model.

Chapter 5

Twelve individual puzzles

ID	FIN characteristic	Nearest known structures	Shared feature	Principal difference
C01	strict oscillatory-damped kernel	spectral graph theory, graph signal processing, band-pass kernels	radial filter and Fourier spectrum	no self-generated continuum or physical interpretation
C02	legacy intermediate bridge profile	screened Green functions, oscillatory propagators	oscillation, damping, amplitude	legacy roles cannot be transferred to strict without a theorem
C03	positive Laplacian A	resistor networks, Markov processes, Dirichlet forms	positivity, constant preservation, gradient energy	time and units are dimensionless
C04	duality e^{-itA} and e^{-tA}	quantum walks, heat kernels, Wick-like continuation	common functional calculus	common generator does not imply operationally identical processes
C05	Green function, resolvent, action	Gaussian fields, inverse field theory, Yukawa-like propagation	operator inverse and quadratic action	interactions, measure, and action scale are additional data
C06	orientation cocycle and chiral receivers	cohomology, holonomy, topological phases, asymmetry resources	sign carrier and reflection transformation	the pair $\pm$ exists, but no sign is selected
C07	fractal Schur compression	Kron reduction, multi-grid, RG, tensor networks	elimination of degrees of freedom	the strict family is not closed under the audited Schur map
C08	adaptive operator and memory	neural operators, reservoir computing, adaptive control	feedback and memory state	learning law and target are not derived from strict
C09	information and geometry	diffusion maps, Fisher geometry, optimal transport	metrics from distributions and semigroups	internal metric is not an SI metre
C10	state–clock–instrument–record	process tensors, causal models, system identification	interventions and multitime records	concrete apparatus remains external
C11	missing dimensional scale	dimensional analysis, renormalization, calibration	scale orbit and need for a standard	no mass, energy, or SI time without an anchor
C12	missing selector	spontaneous symmetry breaking, reference frames, resource theory	multiple branches and asymmetry cost	bifurcation gives branches, not a canonical choice

Chapter 6

Multi-puzzle assemblies

6.1 Spectral core: C01 + C03 + C04

This is the assembly closest to a finite graph transport theory. [\[Proven\]](#)

- W defines connectivity geometry.
- $A = sI - W$ defines Dirichlet energy.
- e^{-itA} produces coherent propagation.
- e^{-tA} produces relaxation/diffusion.

The equivalence is exact at the level of functional calculus, but not at the level of experimental records.

6.2 Propagator inverted into an action: C01 + C03 + C05

If $G = (L + m^2 I)^{-1}$, then the minimal quadratic action

$$S[\phi] = \frac{1}{2} \phi^\top (L + m^2 I) \phi - J^\top \phi$$

has stationary equation $(L + m^2 I)\phi = J$ and response $\phi = GJ$. [\[Proven\]](#)

This correctly reconstructs an action from a propagator. It is not a unique fundamental law, because interactions, measures, and sectors invisible in the two-point Green function may be added.

6.3 Chirality as a resource: C06 + C12

The closest framework is the resource theory of asymmetry. Reflection-covariant operations cannot create asymmetry from a reflection-symmetric state. The prepared phase ϕ is a resource; C_χ is a receiver. [\[Proven\]](#)

This explains why the simulator can read a sign but cannot discharge QW-2191.

6.4 Information as geometry: C04 + C09

The heat semigroup defines diffusion distances and multiscale spectral coordinates, closely resembling diffusion maps. [\[Strong evidence\]](#)

The anti-analogy is essential: P_t can generate a **dimensionless** geometry, but not a metre, second, or energy without calibration.

6.5 Operational observer: C04 + C10

The same A does not determine the observed process. One also needs

$$(\mathcal{H}, \rho, \mathcal{E}_t, \tau, \mathcal{P}, \mathcal{I}, \mathcal{E}, \mathcal{A}, \mathcal{R}, \mathcal{C}, \mathcal{D}).$$

State, intervention, environment, and recording determine whether the record appears wave-like, diffusive, or history dependent. **[Proven]**

6.6 Static compression: C05 + C07

Schur/Kron reduction yields the exact boundary response at a fixed resolvent parameter. This is an equivalence of block algebra, not proof of strict self-similarity. **[Proven]**

The previously audited failure of the strict family to close under the Schur map remains in force.

6.7 Dynamic compression: C04 + C05 + C07 + C10

This is the richest new assembly. When some nodes are invisible, the exact observed process is not generated by one static Schur complement. The frequency-dependent $\Sigma_E(z)$ and a memory kernel appear instead. **[Proven]**

6.8 Fractality plus memory: C07 + C08

If successive layers are eliminated, each contributes its own self-energy and memory kernel, producing a hierarchy

$$\Sigma^{(1)}(z), \Sigma^{(2)}(z), \dots$$

This may resemble continued fractions, multiscale impedance, or hierarchical memory models. **[Moderate evidence]**

No result proves that the hierarchy reproduces the strict kernel or exponent $9/5$.

6.9 Adaptation plus a process with memory: C08 + C10

An adaptive operator learning from process records resembles system identification with memory and neural operators. **[Moderate evidence]**

The model may learn the apparatus rather than FIN. Permutation controls, hold-out tests, and identifiability analysis are essential.

6.10 Two independent obstructions: C11 + C12

The missing scale and missing orientation are different torsors:

- scale carries an action of $\mathbb{R}_{>0}$;
- orientation carries an action of $\mathbb{Z}_2$ or a relevant automorphism group.

A positive scalar cannot select a sign, and a pseudoscalar cannot generate a unit of length. **[Proven]**

This excludes the shortcut in which one "informational constant" is expected to generate both dimensional units and a selector.

6.11 Legacy + Green + strict: C02 + C05 + C01

Legacy may be viewed as an intermediate propagator profile and strict as a later enrichment. **[Moderate evidence]**

This does not license silent substitution of strict for legacy, transfer of legacy physical roles, or treatment of amplitude/damping as a complete completion map.

6.12 Spectral action: C03 + C05 + C09

The trace $\text{Tr } f(A/\Lambda)$ resembles the spectral-action principle, but A alone is not a complete spectral triple and FIN does not generate Λ . **[Moderate evidence]**

This is a useful anti-analogy: the existing theory identifies the additional objects required—an algebra, representation, Dirac-type operator, grading/real structure, and scale.

Chapter 7

New working theorem: dynamic Schur reduction generates memory

Partition the space into observed nodes E and hidden nodes H :

$$A = \begin{pmatrix} A_{EE} & A_{EH} \\ A_{HE} & A_{HH} \end{pmatrix}.$$

7.1 Theorem 1 — exact resolvent block

For $z > 0$,

$$\left[(zI + A)^{-1} \right]_{EE} = \left[zI + A_{EE} - A_{EH}(zI + A_{HH})^{-1}A_{HE} \right]^{-1}.$$

Therefore,

$$\Sigma_E(z) = A_{EH}(zI + A_{HH})^{-1}A_{HE}.$$

Proof. This is the block-matrix inverse formula using the Schur complement. [\[Proven\]](#)

7.2 Theorem 2 — no single exact static reduced generator

For $z > 0$,

$$\Sigma'_E(z) = -A_{EH}(zI + A_{HH})^{-2}A_{HE} \preceq 0.$$

If coupling to the hidden sector is nonzero in an active direction, then $\Sigma'_E(z) \neq 0$. Consequently, no constant matrix B can satisfy

$$\left[(zI + A)^{-1} \right]_{EE} = (zI + B)^{-1}$$

for every $z > 0$.

Proof. The right-hand side would require

$$B = A_{EE} - \Sigma_E(z)$$

to be independent of z , contradicting $\Sigma'_E(z) \neq 0$. **[Proven]**

7.3 Theorem 3 — exact composition defect after projection

Let P select sector E , $Q = I - P^*P$, and let $T(t)$ be either e^{-tA} or e^{-itA} . Then

$$PT(t+s)P^* - PT(t)P^*PT(s)P^* = PT(t)QT(s)P^*.$$

The right-hand side is the amplitude or mass of trajectories that left the observed sector and returned. **[Proven]**

7.4 Time-domain form

For heat dynamics,

$$\dot{x} = -A_{EE}x - A_{EH}h, \quad \dot{h} = -A_{HE}x - A_{HH}h.$$

Exact elimination of h gives

$$\dot{x}(t) = -A_{EE}x(t) + \int_0^t M_E(t-s)x(s)ds - A_{EH}e^{-tA_{HH}}h(0),$$

$$M_E(t) = A_{EH}e^{-tA_{HH}}A_{HE}.$$

This is a memory equation. The "environment" has not been inserted by hand; it arises mathematically from the declared split between visible and hidden degrees of freedom. **[Proven]**

7.5 Result for strict Z_{12}

Choose

$$E = \{0, 2, 4, 6, 8, 10\}, \quad H = \{1, 3, 5, 7, 9, 11\}.$$

Test	Result
static-Schur row-sum residual	2.78×10^{-16}
maximum resolvent-identity residual	3.83×10^{-15}
$\ \Sigma'(0.05)\ _2$	0.919783
$\ \Sigma'(1)\ _2$	0.290955
heat composition defect at $t = 0.5$	0.119015
wave composition defect at $t = 0.5$	0.305427
static-Schur heat error at $t = 1$	0.451941
static-Schur wave error at $t = 1$	0.921176

Test	Result
exact hidden-excursion identity residual	$< 6.3 \times 10^{-16}$

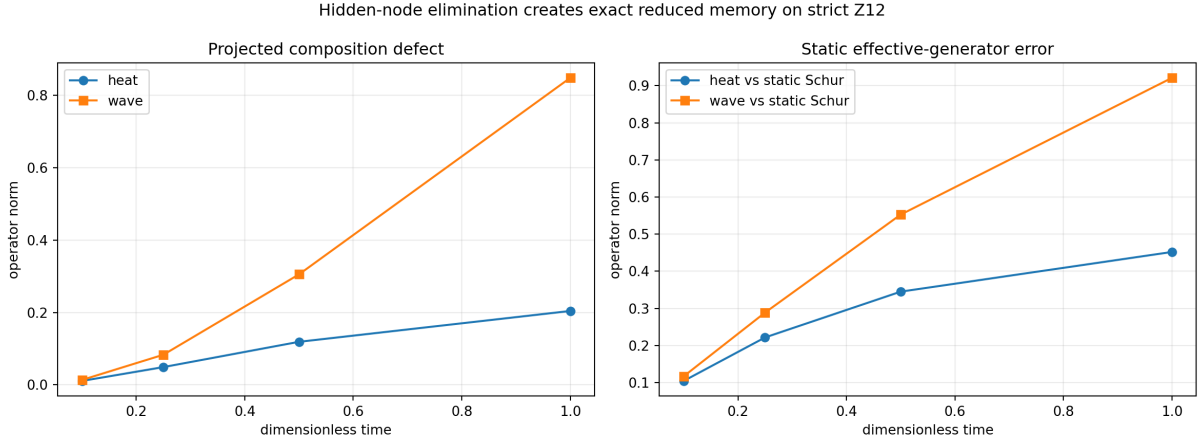


Figure 7.1: Composition defect of reduced propagators and the error incurred by replacing dynamic reduction with one static Schur generator.

7.6 Significance

- **[Proven]** Fractal elimination of degrees of freedom generates memory even when the full dynamics is a homogeneous semigroup or group.
- **[Proven]** Wave and diffusion dynamics share the same block-resolvent object but produce different time records.
- **[Moderate evidence]** $\Sigma_E(z)$ is a strong candidate for the missing mathematical interface between compression, environment, and apparatus.
- **[Refuted]** Static A_{Schur} is not an exact generator of the entire reduced dynamics for the audited partition.
- **[Refuted]** Memory after reduction does not establish a fundamental arrow of time or a conscious observer.

Chapter 8

"Shadow shapes": new intuitions

8.1 The missing object may be a noncommutativity defect

Many unresolved components share the form

reduce first, then evolve $\neq$ evolve first, then reduce.

The defect of this commutation is an observable shadow of hidden degrees of freedom. [\[Strong evidence\]](#)

8.2 The observer is a section of a process, not an added substance

Mathematically, an observer may mean:

- a selected subalgebra of accessible observables;
- a projection P ;
- a set of permitted instruments;
- a rule for recording multitime data.

This does not introduce an "informational layer beneath the nadsoliton." It is an operational section of nadsoliton states. [\[Moderate evidence\]](#)

8.3 C_χ and the chiral bispectrum receive the same resource type

Both are odd under inversion and reverse sign with the preparation. They may belong to a common class of monotone asymmetry receivers. [\[Moderate evidence\]](#) Neither is thereby a sign selector.

8.4 Different observables see different harmonics

The blindness of C_1 when $C_\chi \neq 0$ is a general tomographic lesson. One current or one interference image does not identify the full operator. [\[Proven\]](#)

8.5 Static and dynamic self-similarity are distinct hypotheses

Failure of strict to close under a static Schur map does not rule out a simpler structure for the family $\Sigma^{(n)}(z)$. This is a new, well-typed hypothesis. **[Speculative]**

8.6 Scale can be relative, but an experiment requires an anchor

Spectral ratios are identifiable without an absolute clock; absolute energies and times are not. **[Proven]** A projective fingerprint can be tested first, but a physical theory still requires calibration.

8.7 Information geometry may be real without being spacetime

The heat kernel gives a diffusion metric and scale hierarchy but does not establish Lorentz signature or spacetime geometry. **[Proven]**

8.8 The most defensible architecture still has two additional packages

- W_0 : the strict dimensionless information-operator core;
- an operational/conversion package: preparation, scale, instrument, environment, and record;
- a sector package: an explicit orientation/branch resource.

Dynamic self-energy may enrich the first interface, but it does not remove the other two packages.

Chapter 9

Falsification of analogies

Tempting analogy	Destructive test	Verdict
strict = Yukawa propagator	test whether the inverse and local operator are derived rather than fitted	conditional reconstruction only
duality = observer paradox	same generator but different instruments and records	no mathematical paradox
Schur = RG fixed point	test closure of the strict family	[Refuted] in the audited realization
chiral current = selector	phase inversion gives an equally valid opposite sign	[Refuted] as a source
bifurcation = unique choice	symmetric law produces a pair of branches	[Refuted] without bias
entropy = energy	no temperature, Hamiltonian, or apparatus reset	[Refuted] without additional axioms
diffusion distance = physical length	generator rescaling changes calibration	[Refuted] as an SI metre
process tensor = ontology	depends on accessible interventions and system/environment split	operational description only
$\Sigma(z)$ = new fundamental field	changing the projection changes Σ	[Refuted] as projection independent
graphical resemblance = theory equivalence	no operation- and observable-preserving maps	[Refuted] methodologically

Chapter 10

Twelve subsequent research programs

10.1 Program 243 — Dynamic Schur theorem for arbitrary partitions

Prove conditions for positivity and complete monotonicity of $\Sigma_E(z)$, minimality of the memory realization, and a necessary-and-sufficient criterion for exactly Markovian reduction.

Probability of a valuable result: 0.90.

Success criterion: an if-and-only-if theorem plus tests of every nonisomorphic Z_{12} partition.

10.2 Program 244 — Common analytic self-energy of wave and diffusion

Build one resolvent package and derive its limits

$$z > 0 \quad (\text{heat}), \quad z = \epsilon - i\omega \quad (\text{wave}).$$

Probability: 0.82.

Risk: confusing analytic continuation with physical Wick rotation without axioms.

10.3 Program 245 — Identifiability of the observed/hidden partition

Determine whether distinct E/H splits can generate the same process on E , and construct equivalence classes of minimal realizations.

Probability: 0.80.

10.4 Program 246 — C_χ as a twist-response theorem

Define a flux-twisted family $A(\theta)$ and test

$$C_\chi \stackrel{?}{=} \frac{d}{d\theta} \text{Tr}[\rho A(\theta)] \Big|_{\theta=0}$$

with correct treatment of $d = 6$, gauge covariance, and orientation reversal.

Probability: 0.86.

10.5 Program 247 — Current-harmonic tomography

Measure

$$C_d = \sum_x J(x \rightarrow x + d), \quad d = 1, \dots, 5$$

instead of only C_1 , and determine the minimum preparations and observables required to identify the chiral part of a state.

Probability: 0.84.

10.6 Program 248 — Asymmetry-resource budget

Compare C_χ , the chiral bispectrum, and the earlier $\Lambda(\rho, A)$ as receivers of one resource. Prove bounds of the form

$$|\langle C \rangle_\rho| \leq M_R(\rho) \|C\|_\infty.$$

Probability: 0.78.

Boundary: classification of receivers, not a new selector.

10.7 Program 249 — Fractal cascade of memory

Iterate layer elimination and study

$$\Sigma^{(n+1)}(z) = \mathcal{F}(\Sigma^{(n)}(z)).$$

Seek a fixed point in operator-function space rather than in the existing static-kernel family.

Probability: 0.62.

Reward: high; this would be the proper dynamic version of fractal compression.

10.8 Program 250 — Process-memory test with a complete record

Extend P240–P242 by an intervention at an intermediate time. Distinguish:

- a single semigroup on 12 states;
- a static six-state model;
- the exact six-state process with memory.

Probability: 0.75, laboratory dependent.

10.9 Program 251 — False positives of the spectral fingerprint

Generate broad Laplacian classes and determine how many non-FIN operators pass the projective-spectrum, semigroup, harmonic-current, and prescribed-reduction memory tests.

Probability: 0.88.

10.10 Program 252 — Stability of heat/Fisher geometry under reduction

Compare diffusion distances and the Fisher metric before and after Schur reduction and dynamic reduction.

Probability: 0.73.

Boundary: the result remains dimensionless.

10.11 Program 253 — Can dynamic self-energy generate strict damping?

Treat this as one new bridge atom, not a repetition of a generic legacy-to-strict audit. Test whether elimination of a declared self-similar layer can generate

$$(1 + \beta d^{9/5})^{-1}$$

without encoding 9/5 in the input.

Probability: 0.38.

Stop rule: one explicit ansatz; do not enlarge the family after a no-go.

10.12 Program 254 — Blind hidden-excursion test

Preregister

$$T_E(2\tau) - T_E(\tau)^2 = PT(\tau)QT(\tau)P^*$$

and measure both sides from independent records.

Probability: 0.70.

Value: directly distinguishes static reduction from a process with memory.

Chapter 11

Research priority

Priority	Program	Reason
1	P243	highest theorem confidence and immediate clarification of reduction
2	P246	completes simulator methodology and current definition
3	P247	prevents blindness to all but one harmonic
4	P244	formally joins wave and diffusion after reduction
5	P251	measures specificity of the complete FIN fingerprint
6	P245	determines what can actually be inferred about an "environment"
7	P250	moves the result into an operational multitime process
8	P248	orders chiral receivers without pretending to find a source
9	P252	tests information geometry
10	P254	supplies a falsifiable memory experiment
11	P249	high reward, greater risk
12	P253	most difficult and most vulnerable to target coding

Chapter 12

What this report does not claim

The report does not export:

- a strict selector or discharge of QW-2191;
- a canonical scale of length, time, mass, energy, or action;
- closure of the legacy-to-strict bridge;
- transfer of legacy physical roles;
- L_{total} , the Standard Model, GR, or a ToE;
- evidence that any laboratory realizes FIN;
- evidence that projection-generated memory is fundamental memory in nature.

Chapter 13

Shortest interpretation surviving falsification

The strict FIN core is a finite positive spectral generator whose different functions produce coherent propagation, diffusion, a Green function, and a quadratic action. Chirality belongs to resources of states or operations, not to the radial generator. When some degrees of freedom are hidden—as naturally occurs both in fractal compression and for an observer—the exact shadow of the full theory is not another static kernel but a frequency-dependent self-energy and a process with memory.

This is currently the richest mathematical picture obtained by assembling the puzzles. It is more precise and more testable than the claim that one plot merely "looks like" a familiar phenomenon.

Chapter 14

Comparative literature

1. Dörfler, F.; Bullo, F., *Kron Reduction of Graphs with Applications to Electrical Networks*, <https://arxiv.org/abs/1102.2950>.
2. Lin, Y. T.; Tian, Y.; Anghel, M.; Livescu, D., *Data-driven learning for the Mori–Zwanzig formalism*, <https://arxiv.org/abs/2101.05873>.
3. Pollock, F. A. et al., *Non-Markovian quantum processes: complete framework and efficient characterisation*, <https://arxiv.org/abs/1512.00589>.
4. Marvian, I.; Spekkens, R. W., *Modes of asymmetry*, <https://arxiv.org/abs/1312.0680>.
5. Shuman, D. I. et al., *The Emerging Field of Signal Processing on Graphs*, <https://arxiv.org/abs/1211.0053>.
6. Coifman, R. R.; Lafon, S., *Diffusion maps*, <https://doi.org/10.1016/j.acha.2006.04.006>.
7. Chamseddine, A. H.; Connes, A., *The Spectral Action Principle*, <https://arxiv.org/abs/hep-th/9606001>.

Chapter 15

Reproducible artifacts

- Z12 sim.html — corrected interactive simulator.
- audit_z12_sim.py — independent mathematical audit.
- Z12_sim_methodological_audit.json — results of 15 tests.
- fin_nadsoliton_puzzle_atlas.py — atlas and dynamic-Schur study.
- FIN_Nadsoliton_Puzzle_Atlas_All_Singles.csv — 180 comparisons.
- FIN_Nadsoliton_Puzzle_Atlas_All_Pairs.csv — 990 comparisons.
- FIN_Nadsoliton_Puzzle_Atlas_Results.json — values and candidate ranking.
- FIN_Nadsoliton_Puzzle_Atlas_Figures/ — figures.

Reproduction certificate

```
python3 audit_z12_sim.py
MPLCONFIGDIR=/tmp/matplotlib-puzzle \
python3 fin_nadsoliton_puzzle_atlas.py
```

Expected Z12 audit result: 15/15 tests passed. Atlas scores are search heuristics; operator theorems are verified with separate residuals.